Code: https://github.com/zhurleonid/corporate-risk-management
(`homework_5/hw5_code.ipynb`)

The write-up is below and the code is appended at the end. Part (a), the closed-form price and the two deltas, is derived by hand and attached in front.

Scenario analysis of a delta-hedged spread-squared option

Set-up

Long one unit of the option paying $(S_T^{(1)} - S_T^{(2)})^2$, short $h^{(1)}$ and $h^{(2)}$ shares, with the hedge held fixed over five days, $\Delta = 1/252$:

$$V_t = V(t, S_t^{(1)}, S_t^{(2)}; \sigma_t, \rho_t) - h_t^{(1)} S_t^{(1)} - h_t^{(2)} S_t^{(2)}, \qquad V_{t+5\Delta} = V(t+5\Delta, S_{t+5\Delta}^{(1)}, S_{t+5\Delta}^{(2)}$$

Parameters: $r = 1.32\%$, $\sigma_{1,0} = 22.14\%$, $\sigma_{2,0} = 30.36\%$, $\rho_0 = 0.237$, $T = 0.25$, $S_0^{(1)} = 158.12$, $S_0^{(2)} = 170.33$. The drifts μ_1, μ_2 play no part: the price is risk neutral and the scenarios fix the log returns directly.

(a) The price

> **By hand, attached:** under $\mathbb{Q}$ each $S_T^{(i)}$ is lognormal, so the three terms of $(S_T^{(1)} - S_T^{(2)})^2$ are expectations of exponentials of normals with variances $4\sigma_1^2\tau$, $(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)\tau$ and $4\sigma_2^2\tau$. Discounting by $e^{-r\tau}$ gives

$$V(t, s_1, s_2; \sigma, \rho) = e^{r(T-t)}\left[s_1^2 e^{\sigma_1^2(T-t)} - 2s_1 s_2 e^{\sigma_1\sigma_2\rho(T-t)} + s_2^2 e^{\sigma_2^2(T-t)}\right],$$

$$h^{(1)} = 2e^{r\tau}\left[s_1 e^{\sigma_1^2\tau} - s_2 e^{\sigma_1\sigma_2\rho\tau}\right], \qquad h^{(2)} = 2e^{r\tau}\left[s_2 e^{\sigma_2^2\tau} - s_1 e^{\sigma_1\sigma_2\rho\tau}\right].$$

At $t = 0$:

$$h^{(1)} = -21.9525, \qquad h^{(2)} = 31.2019, \qquad V(0, \cdot) = 921.74, \qquad V_0 = -921.74.$$

V is homogeneous of degree 2 in (s_1, s_2), so $s_1 h^{(1)} + s_2 h^{(2)} = 2V$ and the hedged portfolio is worth exactly $-V$. The hedge is long 22 shares of stock 1 and short 31 of stock 2. Only 149.58 of the option value is the discounted squared spread $e^{rT}(S_0^{(1)} - S_0^{(2)})^2$; the remaining 772.16 is, to first order, $e^{rT}\, T\left(s_1^2\sigma_1^2 - 2\rho\sigma_1\sigma_2 s_1 s_2 + s_2^2\sigma_2^2\right) = 762.91$, the variance of the spread over $[0, T]$. The position is essentially long the variance of the spread, which is what drives the results below.

Pseudocode

```
inputs: r, sigma1_0, sigma2_0, rho_0, T, S1_0, S2_0, Delta
V(t, s1, s2; sig1, sig2, rho) = the closed form from (a), tau
= T − t

hedge at t = 0:   h1 = dV/ds1 (0, S1_0, S2_0; sigma_0,
rho_0),  h2 = dV/ds2 (...)
portfolio:        V_0 = V(0, S1_0, S2_0; sigma_0, rho_0) −
h1*S1_0 − h2*S2_0

grids:  X in {−0.20, −0.10, −0.05, 0.05, 0.10, 0.20}
(each stock)
        k in {0.5, 0.75, 1.25, 1.5, 1.75, 2}
(each vol ratio)
        d_rho in {−0.5, 0, 0.5}
weights: wX(|X|) = 0.5 / 0.75 / 1 for |X| = 0.20 / 0.10 /
0.05
         wK(k)   = 0.5 for k in {0.5, 2}, 1.25 for {0.75,
1.75}, 0.75 for {1.25, 1.5}
         wR(|d_rho|) = 0.5 for 0, 1.25 for 0.5

for each (X1, X2, k1, k2, d_rho) in X x X x K x K x R
(N = 3888):
    S1 = S1_0*exp(X1),  S2 = S2_0*exp(X2)
    sig1 = k1*sigma1_0,  sig2 = k2*sigma2_0,  rho = rho_0 +
d_rho
    V_5 = V(5*Delta, S1, S2; sig1, sig2, rho) − h1*S1 − h2*S2
hedge held fixed
    l_n = −(V_5 − V_0)
    w_n = wX(|X1|) * wX(|X2|) * wK(k1) * wK(k2) * wR(|d_rho|)
    store (scenario, l_n, w_n, w_n*l_n)

(b) report max_n l_n and its scenario
(c) report max_n w_n*l_n, its scenario, and l_n there
```

The sheet's example weight checks out: $X^{(1)} = -0.1$, $X^{(2)} = 0.2$, $k_1 = 2$, $k_2 = 0.5$, $\delta\rho = -0.5$ gives $w = 15/128$.

(b) Worst-case scenario

$$\rho(L_{t+5\Delta}) = \max_n l_n = 1000.36$$

attained at

$$X^{(1)} = X^{(2)} = +0.20, \qquad \frac{\sigma^{(1)}_{t+5\Delta}}{\sigma^{(1)}_t} = \frac{\sigma^{(2)}_{t+5\Delta}}{\sigma^{(2)}_t} = 0.5, \qquad \rho_{t+5\Delta} - \rho_t = +0.5.$$

The loss exceeds the option's whole value at $t = 0$. The eight largest losses:

$X^{(1)}$	$X^{(2)}$	k_1	k_2	$\delta\rho$	ℓ_n	w_n
$+0.20$	$+0.20$	0.50	0.50	$+0.5$	1000.36	$5/64$
$+0.20$	$+0.20$	0.75	0.50	$+0.5$	983.14	$25/128$
$+0.20$	$+0.20$	0.75	0.75	$+0.5$	866.08	$125/256$
$+0.10$	$+0.10$	0.50	0.50	$+0.5$	845.82	$45/256$
$+0.20$	$+0.20$	0.50	0.50	0.0	844.20	$1/32$
$+0.10$	$+0.10$	0.75	0.50	$+0.5$	831.73	0.4395
$+0.20$	$+0.20$	0.50	0.75	$+0.5$	825.35	$25/128$
$+0.20$	$+0.20$	1.25	0.50	$+0.5$	787.04	$15/128$

Why this scenario. Splitting the loss into the price move at the old parameters and the parameter change at the new prices gives $47.59 + 952.77$. The price move itself barely matters: the hedge removes the first-order exposure, and the option is convex in the spread, so a pure price move at fixed parameters loses at most 66 on the whole grid and gains up to 4390 when the two stocks move apart. Almost all of the loss is the fall in spread variance: both volatilities halve and the correlation rises to 0.737, which cuts $s_1^2\sigma_1^2 - 2\rho\sigma_1\sigma_2 s_1 s_2 + s_2^2\sigma_2^2$ by the most the grid allows. Both stocks up 20% comes in because this variance exposure scales with s_1^2 and s_2^2, so higher prices make the same parameter shock cost more, while the spread itself hardly changes when the two move together. The variance approximation of the parameter part is 941.52, within 1.2% of the exact 952.77.

(c) Weighted scenario

$$\rho(L_{t+5\Delta}) = \max_n w_n l_n = 1313.79$$

attained at

$$X^{(1)} = X^{(2)} = +0.05, \qquad \frac{\sigma^{(1)}_{t+5\Delta}}{\sigma^{(1)}_t} = \frac{\sigma^{(2)}_{t+5\Delta}}{\sigma^{(2)}_t} = 0.75, \qquad \rho_{t+5\Delta} - \rho_t = +0.5,$$

with weight $w_n = 1 \times 1 \times 1.25 \times 1.25 \times 1.25 = 125/64 = 1.9531$. **The portfolio loss at this scenario is** $\ell_n = 672.66$, which is 67.2% of the unweighted worst case. The eight largest weighted losses:

$X^{(1)}$	$X^{(2)}$	k_1	k_2	$\delta\rho$	ℓ_n	w_n	$w_n\ell_n$
$+0.05$	$+0.05$	0.75	0.75	$+0.5$	672.66	1.9531	1313.79
-0.05	-0.05	0.75	0.75	$+0.5$	550.52	1.9531	1075.23
$+0.10$	$+0.05$	0.75	0.75	$+0.5$	633.31	1.4648	927.70
$+0.05$	$+0.10$	0.75	0.75	$+0.5$	617.26	1.4648	904.19
$+0.10$	$+0.10$	0.75	0.75	$+0.5$	735.89	1.0986	808.47
$+0.05$	$+0.05$	1.75	0.75	$+0.5$	390.88	1.9531	763.43
$+0.05$	$+0.05$	1.25	0.75	$+0.5$	613.70	1.1719	719.18
-0.10	-0.05	0.75	0.75	$+0.5$	485.07	1.4648	710.55

The weights move the measure from the extreme corner of the grid to a moderate scenario: a small common move up, volatilities down a quarter, correlation up. $125/64$ is the largest weight on the grid, shared by 32 scenarios, and the (c) scenario is the one of those 32 with the largest loss. The weighting discounts the (b) scenario by $5/64$ because halving both volatilities and a 20% move in each stock are all treated as unlikely over five days, whereas -25% in volatility and $+0.5$ in correlation carry weight above one. The direction of the loss is the same as in (b): 66.31 from the price move and 606.35 from the parameter change (variance approximation 599.32). Note that $w_n\ell_n = 1313.79$ is larger than any actual loss on the grid, since the weight exceeds one; the number to compare with (b) is the loss at the scenario, 672.66 against 1000.36.

Appendix: source code

The code below produces every figure and number above.

In [1]:
```
import itertools
import numpy as np
from fractions import Fraction

np.set_printoptions(suppress=True)
```

In [2]:
```
r = 0.0132
sigma1_0, sigma2_0, rho_0 = 0.2214, 0.3036, 0.237
T = 0.25
S1_0, S2_0 = 158.12, 170.33
Delta = 1 / 252
h = 5 * Delta                          # 5-day horizon, the revaluation date

# mu1 = 0.15475 and mu2 = 0.18312 are given on the sheet but play no part:
# the price is risk neutral and the scenarios fix the log returns directly.
```

In [3]:
```
def V(t, s1, s2, sig1, sig2, rho):
    """Price at time t of the payoff (S_T^(1) - S_T^(2))^2."""
    tau = T - t
    return np.exp(r * tau) * (s1**2 * np.exp(sig1**2 * tau)
                              - 2 * s1 * s2 * np.exp(sig1 * sig2 * rho * tau
```

```
                                + s2**2 * np.exp(sig2**2 * tau))


def deltas(t, s1, s2, sig1, sig2, rho):
    """h^(1) = dV/ds1, h^(2) = dV/ds2."""
    tau = T - t
    h1 = 2 * np.exp(r * tau) * (s1 * np.exp(sig1**2 * tau) - s2 * np.exp(sig
    h2 = 2 * np.exp(r * tau) * (s2 * np.exp(sig2**2 * tau) - s1 * np.exp(sig
    return h1, h2


# finite-difference check of the deltas
eps = 1e-5
fd1 = (V(0, S1_0 + eps, S2_0, sigma1_0, sigma2_0, rho_0) - V(0, S1_0 - eps,
fd2 = (V(0, S1_0, S2_0 + eps, sigma1_0, sigma2_0, rho_0) - V(0, S1_0, S2_0 -
print('finite-difference deltas: %.4f  %.4f' % (fd1, fd2))
```

```
finite-difference deltas: -21.9525  31.2019
```

In [4]:

```
h1, h2 = deltas(0, S1_0, S2_0, sigma1_0, sigma2_0, rho_0)
V_opt0 = V(0, S1_0, S2_0, sigma1_0, sigma2_0, rho_0)
V_0 = V_opt0 - h1 * S1_0 - h2 * S2_0

print('h1 = %.4f   h2 = %.4f' % (h1, h2))
print('option value V(0) = %.4f   portfolio V_0 = %.4f' % (V_opt0, V_0))
print('h1*S1 + h2*S2 = %.4f  (= 2 V(0) by homogeneity)' % (h1 * S1_0 + h2 *

# where the option value comes from: intrinsic part vs the variance part
intrinsic = np.exp(r * T) * (S1_0 - S2_0)**2
print('e^{rT}(S1-S2)^2 = %.2f   rest = %.2f' % (intrinsic, V_opt0 - intrinsi
spread_var = sigma1_0**2 * S1_0**2 - 2 * rho_0 * sigma1_0 * sigma2_0 * S1_0
print('first-order approximation of the rest, e^{rT} T (s1^2 sig1^2 - 2 rho
      % (np.exp(r * T) * T * spread_var))
```

```
h1 = -21.9525   h2 = 31.2019
option value V(0) = 921.7399   portfolio V_0 = -921.7399
h1*S1 + h2*S2 = 1843.4797  (= 2 V(0) by homogeneity)
e^{rT}(S1-S2)^2 = 149.58   rest = 772.16
first-order approximation of the rest, e^{rT} T (s1^2 sig1^2 - 2 rho sig1 si
g2 s1 s2 + s2^2 sig2^2) = 762.91
```

In [5]:
```
X_grid = [-0.20, -0.10, -0.05, 0.05, 0.10, 0.20]       # X^(i)_{t+5D}, i = 1
K_grid = [0.5, 0.75, 1.25, 1.5, 1.75, 2.0]              # sigma^(i)_{t+5D} /
R_grid = [-0.5, 0.0, 0.5]                               # rho_{t+5D} - rho_t

wX = {0.20: 0.5, 0.10: 0.75, 0.05: 1.0}                                  # ke
wK = {0.5: 0.5, 2.0: 0.5, 0.75: 1.25, 1.75: 1.25, 1.25: 0.75, 1.5: 0.75}
wR = {0.0: 0.5, 0.5: 1.25}                                               # ke


def weight(x1, x2, k1, k2, dr):
    return wX[abs(x1)] * wX[abs(x2)] * wK[k1] * wK[k2] * wR[abs(dr)]


# the sheet's example: X1 = -0.1, X2 = 0.2, k1 = 2, k2 = 0.5, d rho = -0.5 h
w_ex = weight(-0.1, 0.2, 2.0, 0.5, -0.5)
print('example weight = %.6f = %s' % (w_ex, Fraction(w_ex).limit_denominator
```

```
example weight = 0.117188 = 15/128
```

In [6]:
```
results = []
for x1, x2, k1, k2, dr in itertools.product(X_grid, X_grid, K_grid, K_grid,
    S1 = S1_0 * np.exp(x1)
    S2 = S2_0 * np.exp(x2)
    sig1, sig2, rho = k1 * sigma1_0, k2 * sigma2_0, rho_0 + dr
    V_h = V(h, S1, S2, sig1, sig2, rho) - h1 * S1 - h2 * S2     # same h1, h
    loss = -(V_h - V_0)
    w = weight(x1, x2, k1, k2, dr)
    results.append(dict(X1=x1, X2=x2, k1=k1, k2=k2, drho=dr, loss=loss, w=w,

losses = np.array([s['loss'] for s in results])
print('N = %d scenarios' % len(results))
print('loss: min %.2f   median %.2f   max %.2f   share with a loss > 0: %.1f
      % (losses.min(), np.median(losses), losses.max(), 100 * (losses > 0).m


def show(tag, s):
    print('%s: X1 = %+.2f, X2 = %+.2f, sigma1/sigma1_0 = %.2f, sigma2/sigma2
          % (tag, s['X1'], s['X2'], s['k1'], s['k2'], s['drho']))
    print('    loss = %.4f   weight = %.6f (= %s)   w x loss = %.4f'
          % (s['loss'], s['w'], Fraction(s['w']).limit_denominator(1000), s[


def table(rows, key):
    print('  %6s %6s %6s %6s %6s %10s %8s %10s' % ('X1', 'X2', 'k1', 'k2', '
    for s in sorted(rows, key=lambda s: -s[key])[:8]:
        print('  %+6.2f %+6.2f %6.2f %6.2f %+6.1f %10.2f %8.4f %10.2f'
              % (s['X1'], s['X2'], s['k1'], s['k2'], s['drho'], s['loss'], s
```

```
N = 3888 scenarios
loss: min -8959.53   median -1382.49   max 1000.36   share with a loss > 0:
12.1%
```

In [7]:
```
worst = max(results, key=lambda s: s['loss'])
show('(b) max loss', worst)
print('\neight largest losses')
table(results, 'loss')
```

```
(b) max loss: X1 = +0.20, X2 = +0.20, sigma1/sigma1_0 = 0.50, sigma2/sigma2_
0 = 0.50, d rho = +0.5
    loss = 1000.3617   weight = 0.078125 (= 5/64)   w x loss = 78.1533

eight largest losses
      X1     X2     k1     k2   drho       loss        w     w*loss
   +0.20  +0.20   0.50   0.50   +0.5    1000.36   0.0781      78.15
   +0.20  +0.20   0.75   0.50   +0.5     983.14   0.1953     192.02
   +0.20  +0.20   0.75   0.75   +0.5     866.08   0.4883     422.89
   +0.10  +0.10   0.50   0.50   +0.5     845.82   0.1758     148.68
   +0.20  +0.20   0.50   0.50   +0.0     844.20   0.0312      26.38
   +0.10  +0.10   0.75   0.50   +0.5     831.73   0.4395     365.50
   +0.20  +0.20   0.50   0.75   +0.5     825.35   0.1953     161.20
   +0.20  +0.20   1.25   0.50   +0.5     787.04   0.1172      92.23
```

In [8]:
```
worst_w = max(results, key=lambda s: s['wloss'])
show('(c) max weighted loss', worst_w)
print('\neight largest weighted losses')
table(results, 'wloss')

w_max = max(s['w'] for s in results)
print('\nlargest weight on the grid = %.6f = %s, attained by %d scenarios'
      % (w_max, Fraction(w_max).limit_denominator(1000), sum(s['w'] == w_max
print('the (c) scenario has the largest weight: %s' % (worst_w['w'] == w_max
print('its loss is %.1f%% of the (b) worst case' % (100 * worst_w['loss'] /
```

```
(c) max weighted loss: X1 = +0.05, X2 = +0.05, sigma1/sigma1_0 = 0.75, sigma
2/sigma2_0 = 0.75, d rho = +0.5
    loss = 672.6584   weight = 1.953125 (= 125/64)   w x loss = 1313.7859

eight largest weighted losses
      X1     X2     k1     k2   drho       loss        w     w*loss
   +0.05  +0.05   0.75   0.75   +0.5     672.66   1.9531    1313.79
   -0.05  -0.05   0.75   0.75   +0.5     550.52   1.9531    1075.23
   +0.10  +0.05   0.75   0.75   +0.5     633.31   1.4648     927.70
   +0.05  +0.10   0.75   0.75   +0.5     617.26   1.4648     904.19
   +0.10  +0.10   0.75   0.75   +0.5     735.89   1.0986     808.47
   +0.05  +0.05   1.75   0.75   +0.5     390.88   1.9531     763.43
   +0.05  +0.05   1.25   0.75   +0.5     613.70   1.1719     719.18
   -0.10  -0.05   0.75   0.75   +0.5     485.07   1.4648     710.55

largest weight on the grid = 1.953125 = 125/64, attained by 32 scenarios
the (c) scenario has the largest weight: True
its loss is 67.2% of the (b) worst case
```

In [9]:
```
def decompose(s):
    S1, S2 = S1_0 * np.exp(s['X1']), S2_0 * np.exp(s['X2'])
    sig1, sig2, rho = s['k1'] * sigma1_0, s['k2'] * sigma2_0, rho_0 + s['drh
    port = lambda a, b, c: V(h, S1, S2, a, b, c) - h1 * S1 - h2 * S2
    l_move = -(port(sigma1_0, sigma2_0, rho_0) - V_0)              # prices
    l_param = -(port(sig1, sig2, rho) - port(sigma1_0, sigma2_0, rho_0))   #
    tau = T - h
    var_old = sigma1_0**2 * S1**2 - 2 * rho_0 * sigma1_0 * sigma2_0 * S1 * S
    var_new = sig1**2 * S1**2 - 2 * rho * sig1 * sig2 * S1 * S2 + sig2**2 *
    approx = -np.exp(r * tau) * tau * (var_new - var_old)
    print('  price move only: %8.2f   parameter change: %8.2f   total: %8.2f
```

```
          % (l_move, l_param, l_move + l_param, approx))


print('(b) scenario'); decompose(worst)
print('(c) scenario'); decompose(worst_w)

# the price move alone, over the whole X grid at the t = 0 parameters: the h
base = [s for s in results if s['k1'] == 1.25 and s['k2'] == 1.25 and s['drh
print('\nloss from the price move alone (parameters unchanged), X1 down the
print('        ' + ''.join('%8.2f' % x for x in X_grid))
for x1 in X_grid:
    S1 = S1_0 * np.exp(x1)
    row = []
    for x2 in X_grid:
        S2 = S2_0 * np.exp(x2)
        row.append(-(V(h, S1, S2, sigma1_0, sigma2_0, rho_0) - h1 * S1 - h2
    print('  %+.2f ' % x1 + ''.join('%8.2f' % v for v in row))
```

```
(b) scenario
  price move only:    47.59   parameter change:   952.77   total:  1000.36
(variance approximation of the parameter part:   941.52)
(c) scenario
  price move only:    66.31   parameter change:   606.35   total:   672.66
(variance approximation of the parameter part:   599.32)

loss from the price move alone (parameters unchanged), X1 down the rows, X2
across
           -0.20   -0.10   -0.05    0.05    0.10    0.20
  -0.20    11.40 -119.47 -372.70-1354.21-2129.66-4389.67
  -0.10  -228.62   42.57    5.96 -508.42-1032.19-2749.47
  -0.05  -513.84  -26.03   54.08 -208.63 -596.80-2021.68
  +0.05 -1500.44 -545.50 -213.72   66.31  -29.46 -823.78
  +0.10 -2242.43-1035.82 -568.44    3.99   65.76 -388.82
  +0.20 -4344.05-2594.70-1834.92 -631.93 -230.43   47.59
```